

SAMPLING DISTRIBUTIONS & THE CLT

new 2.12, 3.1–3.2, 4.1 • old 5.1–5.7

ZONE A · THE FORMULAS + CONDITIONS

1 · RANDOM: START WITH A RANDOM SAMPLE

SAMPLE PROPORTION $\hat{p}$
Population proportion p

$$\mu_{\hat{p}} = p$$

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

SAMPLE MEAN $\bar{x}$
Population mean μ , SD σ

$$\mu_{\bar{x}} = \mu$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

2 · 10%: $n < 0.10 \cdot N$ when sampling without replacement.
Draws are close enough to independent to use these SD formulas.

3 · LARGE COUNTS

The distribution of $\hat{p}$ is approximately normal when
 $np \geq 10$ and $n(1-p) \geq 10$.

3 · NORMAL / CENTRAL LIMIT THEOREM

Normal population $\Rightarrow$ normal $\bar{x}$, for any n .
Otherwise, CLT gives an **approximate** normal shape for large n . The $n \geq 30$ guideline is approximate; strong skew can need a larger sample.

ZONE B · PARAMETERS VS STATISTICS

	Parameter	Statistic
Mean	μ	$\bar{x}$
SD	σ	s
Proportion	p	$\hat{p}$
Slope	β	b

Parameters are Greek or unhatted. Statistics wear a hat or a bar (or use the sample symbols s and b).

Unbiased: the sampling distribution is centered at the parameter.

Variability shrinks with $\sqrt{n}$: **quadrupling n halves the SD.**

ZONE C · SAY IT IN WORDS

"The sampling distribution of _____ is the distribution of that statistic from all possible samples of size $n = \underline{\hspace{1cm}}$ from _____."

One sample gives one statistic; repeated samples give a distribution.

ZONE D · WATCH OUT

KEEP n AND N SEPARATE

n is sample size; N is population size.
Sample size controls these SDs.
Use N for the 10% check.

SD VS ESTIMATED SD

$\sigma_{\bar{x}} = \sigma/\sqrt{n}$ needs population SD σ . With sample SD s , $s/\sqrt{n}$ is an **SE** estimating the SD of $\bar{x}$. See P13.